

Day-1

Task-1

- **Computer won't turn on:** Check the power supply's output voltage to determine whether the PSU is delivering the correct power.
- **Laptop battery not charging:** Measure the battery or charging circuit voltage to identify whether power is reaching the battery properly.
- **Smartphone won't charge:** Test the USB charging port and cable for voltage and continuity to locate a faulty connection.

Task-2

A POST (Power-On Self-Test) card is a diagnostic tool that slots directly into a motherboard to intercept and display hex codes generated by the BIOS during startup. By reading these codes on the card's digital display, technicians can pinpoint exactly which hardware component (like RAM, CPU, or a power rail) is causing the system to fail before it even reaches the operating system. You can cross-reference the displayed error code with the motherboard manufacturer's manual to immediately identify the culprit.

Task -3

Sensor	Value	Min	Max
RAFTAAR HP 89EB			
Temperatures			
TZ.THRM	55.0 °C	55.0 °C	58.0 °C
Utilization			
Physical Memory Load	67.0 %	67.0 %	67.0 %
Available Physical Memory	2.4 GB	2.4 GB	2.4 GB
Virtual Memory Load	72.4 %	72.2 %	72.6 %
Available Virtual Memory	3.4 GB	3.4 GB	3.4 GB
AMD Ryzen 5 5600H			
Voltages			
VID (Max)	1.231 V	0.719 V	1.237 V
CPU VDD	1.294 V	0.725 V	1.300 V
SoC VDD	0.925 V	0.681 V	0.931 V
Temperatures			
Cores (Max)	48.7 °C	48.3 °C	54.0 °C
Package	54.3 °C	53.4 °C	60.5 °C
L3 Cache	48.1 °C	48.1 °C	51.3 °C
GFX	47.5 °C	47.5 °C	49.0 °C
Powers			
Package	4.48 W	4.03 W	13.13 W
Cores	1.21 W	0.52 W	5.89 W
L3 Cache	0.41 W	0.37 W	0.78 W
Currents			
CPU VDD	13.51 A	1.04 A	28.06 A
SoC VDD	8.31 A	1.04 A	13.51 A
Levels			
Energy Performance	49.8 %	49.8 %	49.8 %
Clocks			
Cores (Max)	4039.7 MHz	1974.7 MHz	4138.3 MHz
CPU BCLK	99.7 MHz	99.6 MHz	99.8 MHz
Memory Controller	1595.9 MHz	1593.9 MHz	1596.7 MHz
Memory	1593.9 MHz	1593.9 MHz	1596.2 MHz
Utilization			
Processor	5.3 %	2.0 %	8.4 %
WD PC SN810 SDPCPNRY-512G-1006 (23041N808027)			
Temperatures			
Assembly	43.0 °C	43.0 °C	43.0 °C
Counters			
Power-On-Hours	1639 hrs	1639 hrs	1639 hrs
Power Cycle Count	3725	3725	3725
Health Status	93 %	93 %	93 %
Utilization			
Used Space (C)	89.9 %	89.9 %	89.9 %

Task-4

Tool	What it checks	Example Problem Detected
Multimeter	Raw electrical parameters (DC/AC voltage, resistance, continuity, and current) directly on PSU cables, VRMs, capacitors, and motherboard traces.	A dead Power Supply Unit (PSU) failing to output +12V DC on the 8-pin CPU connector, or a short circuit across a motherboard power rail.
POST Card	Low-level BIOS initialization sequences and Port 80h hexadecimal error codes via expansion buses (PCI, ISA) before video output or OS boot.	A system hanging in a black-screen state due to unseated or defective RAM modules (e.g., halting at early memory initialization code 53).
Software Diagnostics (e.g., <i>MemTest86</i> , <i>HWMonitor</i> , <i>CrystalDiskInfo</i>)	Live hardware telemetry, memory address stability, S.M.A.R.T. disk health attributes, and thermal stats after or during system execution.	A solid-state drive accumulating critical reallocated sector errors, or a CPU severely thermal throttling under load due to dried thermal paste.